

Singleton condition tested against all irreducible representations

`MathlibAnnex.Analysis.CStarAlgebra.Representation.IsSingletonIrreducibleModelAmongNonUnital`

def

Defines the unital singleton condition while allowing the quantified representations to be presented without a unit equation.

Statement

Let A be a unital complex C^* -algebra, let H be a complex Hilbert space, and let $\pi : A \rightarrow \mathcal{B}(H)$ be a unital $*$ -representation. The predicate below requires π to be nonzero irreducible and compares it with every nonzero irreducible $*$ -representation $\rho : A \rightarrow \mathcal{B}(K)$, even when the data type used for ρ does not initially record $\rho(1_A) = I_K$.

Definition

For every complete complex Hilbert space K and every $*$ -representation $\rho : A \rightarrow \mathcal{B}(K)$ that is nonzero irreducible, irreducibility forces

$$\rho(1_A) = I_K.$$

Indeed, $\rho(1_A)$ is an idempotent whose range is a nonzero closed reducing subspace, hence all of K . The source expression `rho.toUnital` therefore packages the same operator-valued map as a unital representation. The required conclusion is a unitary $U : H \rightarrow K$ satisfying

$$U\pi(a) = \rho(a)U \quad (a \in A).$$

Assumptions

No separability or faithfulness assumption is included. The distinction is between a unital algebra and whether unit preservation is recorded in the representation interface; it is not a claim that A or ρ lacks a unit.

Conclusion

The predicate says that π represents the unique unitary-equivalence class of nonzero irreducible $*$ -representations of the unital algebra A , with the same operator maps whether or not the input interface initially records the unit equation.

The two implications in Both directions of the singleton comparison identify this condition with the corresponding comparison restricted to unital representations: one direction regards the same unital representation through the interface that does not require a unit equation, the other uses the forced equation just proved. These remain distinct declared predicates and distinct Cards.

Main citations

- Exact definition and context
- A nonzero irreducible representation of a unital algebra preserves its unit
- Both directions of the singleton comparison

Lean source signature (exact)

The complete declaration below is a separate exact source excerpt; the original header record is retained with the manuscript.

```
-- Raw singleton-spectrum hypothesis for a unital algebra, with every
ordinary nonzero irreducible representation included and no faithfulness
built into the definition. -/
def IsSingletonIrreducibleModelAmongNonUnital
  (pi : Representation A H) : Prop :=
  pi.IsIrreducible ∧
  ∀ (K : Type w) [NormedAddCommGroup K] [InnerProductSpace ℂ K]
    [CompleteSpace K] (rho : NonUnitalRepresentation (A := A) (H := K)),
    ∀ hrho : rho.IsIrreducible,
      pi.UnitaryEquivalent (rho.toUnital hrho)
```

In the source	Mathematical meaning
(pi : Representation A H)	The specified unital $\ast$ -representation $\pi : A \rightarrow \mathcal{B}(H)$.
pi.IsIrreducible ∧	π is nonzero irreducible, and the following universal comparison also holds.
∀ (K : Type w) [NormedAddCommGroup K] [InnerProductSpace ℂ K] [CompleteSpace K]	Every complete complex Hilbert space K ; no separability restriction is imposed.
(rho : NonUnitalRepresentation (A := A) (H := K))	A $\ast$ -representation $\rho : A \rightarrow \mathcal{B}(K)$ for which the data type does not require a unit-preservation equation.
∀ hrho : rho.IsIrreducible	Assume that this same ρ is nonzero irreducible.
rho.toUnital hrho	The same map $a \mapsto \rho(a)$, now equipped with the derived equality $\rho(1_A) = I_K$.
pi.UnitaryEquivalent (rho.toUnital hrho)	There is a unitary $U : H \rightarrow K$ such that $U\pi(a) = \rho(a)U$ for every $a \in A$.

Exact source and proof.

Earlier published Card and PDF

▼ **Exact content identity**

Declaration: `MathlibAnnex.Analysis.CStarAlgebra.Representation.IsSingletonIrreducibleModelAmongNonUnital`

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